

Faithfulness, Worked Out by Hand

The metric that catches hallucination in RAG — it breaks the answer into atomic claims and asks what fraction are actually supported by the retrieved context. Computed claim by claim on a worked example.

What this document does. It defines faithfulness as the share of an answer’s claims that the retrieved context can back up, extracts the claims from a sample answer, verifies each against the context, and computes the score. It distinguishes faithfulness (grounding) from relevance (usefulness) so the two are not confused. Numbers from Appendix A.

One of four RAGAS metrics: faithfulness (#25), answer relevance (#26), context precision (#27), context recall (#28). Together they localise where a RAG pipeline fails.

1 What it measures

A RAG answer is *faithful* if everything it asserts can be traced back to the retrieved context — no invented facts, no unsupported leaps. Faithfulness ignores whether the answer is *helpful* or even *correct in the world*; it asks only: *is the answer grounded in the documents you gave the model?*

$$\text{Faithfulness} = \frac{\#\{\text{claims in the answer supported by the context}\}}{\#\{\text{total claims in the answer}\}} \in [0, 1].$$

Intuition

Think of the model as a witness on the stand who may only testify to what is in the case file (the retrieved context). Faithfulness audits the transcript: break the testimony into individual factual statements and check each one against the file. A statement the file supports is fair; a statement the witness made up — however plausible — is a hallucination. The score is the fraction of testimony that holds up. It does not ask whether the file was the *right* file (that is context recall, #28) or whether the testimony *answered the question* (that is answer relevance, #26).

2 The procedure

1. **Decompose** the answer into atomic claims — each a single, independently checkable assertion.
2. **Verify** each claim against the retrieved context: can it be inferred from the context? (In RAGAS an LLM judge makes this entailment call.)
3. **Divide** supported claims by total claims.

3 Worked by hand

Question. “Tell me about the Eiffel Tower.”

Retrieved context. “The Eiffel Tower is a wrought-iron lattice tower in Paris, France. Completed in 1889, it stands 330 metres tall.”

Generated answer. “The Eiffel Tower is in Paris. It is 330 m tall. It was built in 1889. It is made of aluminium.”

Step 1–2 — decompose into claims and verify each.

claim	supported by context?	evidence
1. It is in Paris	yes	“in Paris, France”
2. It is 330 m tall	yes	“stands 330 metres tall”
3. It was built in 1889	yes	“Completed in 1889”
4. It is made of aluminium	no	context says “wrought-iron”

Step 3 — score.

$$\text{Faithfulness} = \frac{3 \text{ supported}}{4 \text{ total}} = \boxed{0.75}.$$

Three claims are grounded; the fourth (**aluminium**) contradicts the context’s **wrought-iron** — a textbook hallucination. The score 0.75 flags that one quarter of the answer is unsupported.

Mini-example · plausible is not the same as grounded

“Made of aluminium” is a *reasonable-sounding* fact about a metal tower — it is exactly the kind of fluent, confident invention LLMs produce. Faithfulness does not care that it sounds right; it cares that the *context* did not say it (and here contradicts it). This is why faithfulness is the front-line hallucination guard: it penalises the model for asserting anything the documents do not back, no matter how convincing.

4 Faithfulness vs. the other failure modes

- **Low faithfulness** $\Rightarrow$ the *generator* is hallucinating — it had the context but strayed from it. Fix the prompt (demand grounding/citations), the model, or the decoding (lower temperature, #2).
- **High faithfulness, wrong answer** $\Rightarrow$ usually a *retrieval* problem: the model faithfully used context that did not contain the real answer (check context recall, #28). Faithful to the wrong documents is still wrong.
- **Faithfulness ignores relevance**: an answer can be perfectly grounded yet not address the question. That is answer relevance’s job (#26).

Used together, the four RAGAS metrics tell you *which stage* broke — the whole point of measuring them separately.

Equation cheat-sheet

$$\text{Faithfulness} = \frac{\text{supported claims}}{\text{total claims}} \in [0, 1]$$

low $\Rightarrow$ generator hallucinates; grounding only — not correctness, not relevance

Appendix A · Reference sketch

An LLM judge does steps 1-2; the metric is the final ratio.

```
claims = [
    ("in Paris", True), # "in Paris, France"
    ("330 m tall", True), # "stands 330 metres tall"
    ("built in 1889", True), # "Completed in 1889"
    ("made of aluminium", False), # context says "wrought-iron"
]
faithfulness = sum(ok for _, ok in claims) / len(claims) # 3/4 = 0.75
```

Internal learning material. Numbers reproducible from the appendix sketch. v1.0